

RUOYU LI

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NYU M.S. student in Urban Data Science, Research Assistant on the Urban Nighttime Lighting Index (UNLI) team, and Data Analysis and Social Impact Intern at DarkSky International. Advocate for light pollution reduction.

Research Interests: Policy-oriented environmental research; nighttime light remote sensing; spectral image analysis (multi- and hyperspectral); spatiotemporal data analytics; environmental justice; community well-being.

SKILLS

Remote Sensing (Nighttime Light, Multi-/Hyperspectral Imaging, LiDAR) | Spatiotemporal Data Analytics & Visualization | Public Policy Research, Analysis & Evaluation | Causal Inference | Machine Learning & GeoAI | Network Science | Information Science | Web Development

Tools: Python (Jupyter, GeoPandas, Rasterio, scikit-learn, PyArrow, Folium) | R | Google Earth Engine | ArcGIS Pro | QGIS | CloudCompare | Git/GitHub | Conda | HTML/CSS/JavaScript | Leaflet | Tableau

EDUCATION

New York University, *New York, NY* | *Expected May 2026*

- **MS, Urban Data Science**, Center for Urban Science and Progress (CUSP), Tandon School of Engineering
- **Honors:** Tandon's highest merit scholarship | CUSP Experiential Scholars
- **Capstone:** Urban Colocation Intelligence: Mapping Amenity Clusters and Dependency Networks Using Open-Source POI Data (Project Sponsor: Foster + Partners)
- **Selected Activities:** American Society for Photogrammetry and Remote Sensing (ASPRS) NYU Chapter (Founding Treasurer) | NYC Geospatial Information Systems and Mapping Organization (GISMO) (GeoAI Committee) | Tandon Vertically Integrated Projects (VIP) Urban Remote Sensing Team (Hyperspectral Image Analysis) | Public Data Challenge (Finalist) | Global Data Dive (3rd Place) | Tandon Capstone Competition (Selected) | NYU GIS Day 2025 Mapping Competition (2nd Place) | NYC Open Data Week 2026 (Workshop Host)
- **Volunteers:** Climate Week NYC | World Environment Day | NYC School of Data | CityCamp NYC

Columbia University, *New York, NY* | *Jul 2021*

- **Graduate Certificate in Advanced Business**

University of Arizona, *Tucson, AZ* | *May 2019*

- **BS, Mathematics** | **BS, Information Science and Technology**
- **Minors:** Statistics and Data Science | Finance | Computer Science
- **Certificate:** Entrepreneurship and New Venture Development
- **Honors:** Academic Distinction | Dean's List with Distinction | Dean's List | Director's List
- **Selected Activities:** Arizona Chinese Culture Club (Co-founder & VP) | Association of Chinese Students and Scholars (Marketing Director) | McGuire New Venture Development Program (Finalist)

EXPERIENCES

New York University, *New York, NY*

Graduate Research Assistant | *Mar 2026 – Present*

- Part of the Urban Nighttime Lighting Index (UNLI) research project sponsored by the NYU Goddard Impact Award that studies urban nighttime light pollution and its impacts on human health.
- Leading the quantification of urban outdoor lighting at the census tract level using nighttime light satellite imagery and field-collected data; leading the field data collection.
- Analyzing the correlation of outdoor lighting with negative public health outcomes.

DarkSky International, Tucson, AZ

Data Analysis and Social Impact Intern | Feb 2026 – Present

- Working on a foundational assessment of the International Dark Sky Places (IDSP) program's impact through lighting policy compliance analysis, sky-quality submissions, and open data on nighttime brightness.

New York State & Beijing Chapter Leader | Mar 2015 – Present

- Led advocacy efforts for light pollution control in New York State and China.
- Helped establish multiple International Dark Sky Places in China.

Information Technology Manager & Regional Engagement Associate | Oct 2023 – Aug 2024

- Managed IT infrastructure to improve efficiency; provided support for staff and advocates.
- Enhanced regional engagement and developed DarkSky's China strategy.

China Academy of Urban Planning and Design, Shenzhen, China

Intern (Research Center for Urban Lighting Planning and Design) | Nov 2023 – Jan 2024

- Contributed to citywide lighting master plans and participated in public lighting art installation projects.
- Co-authored the *Urban and Rural Dark Sky Places Planning Standards*.

Eco Green Recycle Corp., San Juan, Puerto Rico

Co-President | Aug 2019 – Aug 2023

- Supervised daily operations; employed data-driven strategies to enhance production & financial performance.

National Museum of Mathematics, New York, NY

Intern | Jun 2018 – Aug 2018

- Alternated between roles in administration, museum docent, and retail assistant.

Arizona/NASA Space Grant Consortium, Tucson, AZ

Research Intern (ASCEND! Program) | Sep 2014 – May 2016

- Member of a statewide aerospace student program, design, build, fly, recover, and analyze small payloads with multiple sensors and cameras for launch from high-altitude (~100,000 ft) weather balloons.

National Astronomical Observatories (NAOC), Chinese Academy of Sciences (CAS), Beijing, China

Research Intern | Jun 2015 – Jul 2015

- Participated in a sky survey research project, learned data processing, and studied professional telescope operations on-site.

SELECTED PROJECTS

In progress:

- The Causal Effect of Municipal Lighting Policy on Nighttime Brightness — *Built a spatiotemporal nighttime-brightness dataset from satellite imagery; applied causal inference methods to evaluate policy impacts.*
- Automated Hyperspectral Assessment of Sidewalk Pavement Conditions — *Preprocessed hyperspectral imagery with QGIS and clipped using NYC's planimetric map; trained machine learning models with Python to classify and visualize sidewalk pavement conditions based on spectral signature.*
- Greener NYC: A Spatial Analysis of Urban Green Spaces and Public Health Outcomes — *Computed green space indicators from land cover raster data; performed census-tract-level zonal statistics and spatial correlation analysis on public health open data; mapped disparities.*
- Characterizations of International Dark Sky Communities in the U.S. — *Derived nighttime-light exposure metrics for designated communities; integrated socioeconomic + public health open data; produced ArcGIS StoryMaps to support benchmarking and policy evaluation.*

- Urban Colocation Intelligence: Mapping Amenity Clusters and Dependency Networks (NYU Capstone; Sponsor: Foster + Partners) — *Cleaned and harmonized large-scale POI/mobility data of London; clustered amenities and built dependency networks; packaged methods into a reusable toolkit for urban design workflows.*

Completed:

- Urban nighttime spectral responses and multi-species risk assessment based on SDGSAT-1 glimmer imagery — *Integrated SDGSAT-1 nighttime imagery with field-collected ground-based measurements; built multispectral inversion and urban-scale spectral response index models with robust validation.*
- Light Pollution Indicator Weighted by Urban Development Level — *Analyzed ground-based sky-quality data; linked measures with socioeconomic factors to develop weighted indicators for cross-city comparisons.*
- Hidden Exposure: Personal Air Pollution Monitoring Reveals Gaps in Public Urban Sensing Systems — *Collected personal air quality sensor data; modeled exposure patterns with machine learning; built interactive web maps (Folium/Leaflet) to highlight gaps in public monitoring.*
- Impact of COVID-19 on Urban Mobility: A Spatial Analysis of NYC Citi Bike Ridership Changes — *Built a spatiotemporal ridership dataset; quantified census-tract-level changes via spatial analysis/zonal statistics; contextualized patterns with public health indicators.*
- Map the NYC 2025 Primary & General Mayoral Election at the Election District (ED) Level (2nd Place, 2025 NYU GIS Day Mapping Competition) — *Scraped and processed election night results; built an ED-level interactive map; published a reproducible open-source workflow on GitHub.*
- Google Maps Timeline Data Viewer & Heatmap Visualization — *Built a web-based interactive location-history viewer and a heatmap viewer; released code as an open-source project.*
- Airport Heat Island: Phoenix, Dallas, and NYC — *Computed heat island metrics in Google Earth Engine; compared airport vs surrounding urban areas across cities for working and living conditions evaluation.*
- Awareness of Privately Owned Public Spaces (POPS) — *Evaluated POPS policy, public awareness, and public accessibility; mapped site distribution; synthesized findings into recommendations.*
- Implementing Advanced Catalytic Plastic Recycling Technology in New York City — *Developed a NYC-specific business case analysis; built a location selection model; assessed operational feasibility for implementation.*

SELECTED PUBLICATIONS, PRESENTATIONS & ACTIVITIES

Publications:

- Li, R., Liu, M., Long, T. F., **Li, R.**, Jiao, F., Wang, Z., & Yuan, S. (n.d.). *Urban nighttime spectral responses and multi-species risk assessment based on SDGSAT-1 glimmer imagery* **[Manuscript under review]**.
- **Li, R.** (2022, June 20). Dark Sky State: Arizona, US, *Chinese National Astronomy*, 16(6), 30-33.

Organizing / Planning / Coordinating / Hosting:

- NYC Open Data Week workshop: “Reclaiming the Night — Urban Light Pollution, Solutions, and Open Data,” *New York, NY, 2026*
- Climate Week NYC x SEE the Future Forum: “Entrepreneurship and Coalition-Building for Global Climate Action,” *New York, NY, 2025*
- International Dark Sky Week x NYU: “Mitigating Light Pollution & Rediscovering the Night,” *New York, NY, 2025*
- Shanghai Astronomy Museum Photography Exhibition: “Guarding the Dark Night | Aiming at the Stars,” *Shanghai, China, 2023-24*
- Forum for the Dark and Quiet Sky Protection, International Year of Basic Sciences for Sustainable Development (IYBSSD), *Hebei, China, 2023*
- DarkSky Global Conference, *Virtual, 2021 & 2023*

Media / Interviews:

- Mancall-Bitel, N., & Moran, K. (2025, December 2). New York's Skyline Has a Bold New Look. *The New York Times*.
- Island Stars, 2025
- *News and Science*, Hunan TV, 2024
- Bi-City Biennale of Urbanism/Architecture (UABB) of Shenzhen and Hong Kong, 2022

Invited Talks:

- Safe, Just, and Thriving Lightning Talks, NYU-2026 PIT-UN Tech for Change Hackathon, *New York, NY, 2026*
- Ethical Night Rangers, New York Society for Ethical Culture, *New York, NY, 2025-2026*
- Dark Sky and Territorial Spatial Planning (class), Guangzhou University, *Guangdong, China, 2025*
- Brooklyn Heights Garden Club, *New York, NY, 2025*
- Guangdong-Hong Kong-Macao Greater Bay Area Dark Sky Astronomy Festival, *Guangdong, China, 2024*
- Yangzhou Urban Lighting Management Office, *Jiangsu, China, 2023*
- Ningbo Haishu District Government, *Zhejiang, China, 2023*
- Hangzhou High School, *Zhejiang, China, 2023*
- The Fifth Star Party of the Yangtze River Delta, *Zhejiang, China, 2023*
- Forum for the Dark and Quiet Sky Protection, IYBSSD, *Hebei, China, 2023*
- "Under One Sky" Live Event, UABB of Shenzhen and Hong Kong, *Guangdong, China, 2022*
- Academic Salon, Youth Innovation Promotion Association, NAOC, CAS, *Beijing, China, 2022*
- Student Seminar, National Astronomical Observatories (NAOC), CAS, *Beijing, China, 2015*

Presentations / Posters / Lightning Talks / Showcases:

- NYU GIS Day 2025 Mapping Competition, *New York, NY, 2025*
- CUSP Summer Research Showcase, *New York, NY, 2025*
- Tandon Capstone Showcase, *New York, NY, 2025*
- NYC Open Data Week x GISMO, *New York, NY, 2025*
- CUSP London Global Data Dive Showcase, *New York, NY, 2025*
- DarkSky Global Conference, *Phoenix, AZ, 2015; Tucson, AZ, 2016; Boston, MA, 2017; Salt Lake City, UT, 2018*
- Hack Arizona, *Tucson, AZ, 2016*